Roll No.: | | | | | | | | | | | |

Test Date: 01-11-2026

Aakash
Medical | IIT-JEE | Foundations

I
Booklet Code

Aakash South Zone-B

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A I A T S
ALL INDIA AAKASH TEST SERIES

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for

Medical Entrance Exam - 2028

National Eligibility-cum-Entrance Test (NEET)

TEST No. 1

(XI Studying Students)

(Phase-3 & 4)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

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MM : 720

TEST - I

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer:

1. Which of the following statement among the given options is correct?
   (1) Most precise measurement must be most accurate
   (2) Most accurate measurement must be most precise
   (3) Least precise measurement may be most accurate
   (4) Measurement using a vernier calliper is less precise than measurement using a metre scale

2. The error in measurement of side of a cube is 1%. The error in measurement of its total surface area is
   (1) 1% (2) 2%
   (3) 3% (4) 4%

3. If $h << R$, then approximate value of $\left(1+\frac{h}{R}\right)^{-2}$ is (h and R are non-zero real numbers)
   (1) $\left(1-\frac{2h}{R}\right)$ (2) $\left(1+\frac{2h}{R}\right)$
   (3) $\left(1+\frac{h}{2R}\right)$ (4) $\left(1-\frac{h}{2R}\right)$

4. Which of the following equations is dimensionally incorrect? (symbols have their usual meanings)
   (1) $F=\frac{Gm_1m_2}{R^2}$ (2) $v^2=u^2+as$
   (3) $T=2\pi\sqrt{\frac{l}{g}}$ (4) $s=ut+\frac{1}{2}at^2$

5. The position (x) of a particle at time (t) is given by the relation, $x=\frac{v_0}{\alpha}\left[1-e^{-\alpha t}\right]$. The dimensional formula for $\alpha^2 v_0^3$ will be
   (1) $[M^0L^3T^{-4}]$ (2) $[M^0L^3T^{-3}]$
   (3) $[M^0LT^{-2}]$ (4) $[M^0L^2T^{-3}]$

6. Which of the following is a derived quantity?
   (1) Mass (2) Velocity
   (3) Length (4) Time

7. Unit of power is
   (1) kilowatt (2) kilowatt × hour
   (3) dyne (4) joule

8. If measured values of mass (m), radius (r) and length (l) of a wire are as follows:
   $m = (0.60 \pm 0.003)$ g, $r = (0.50 \pm 0.01)$ cm and $l = (10.00 \pm 0.05)$ cm
   The maximum percentage error in the measurement of density of wire is
   (1) 8% (2) 5%
   (3) 4% (4) 7%

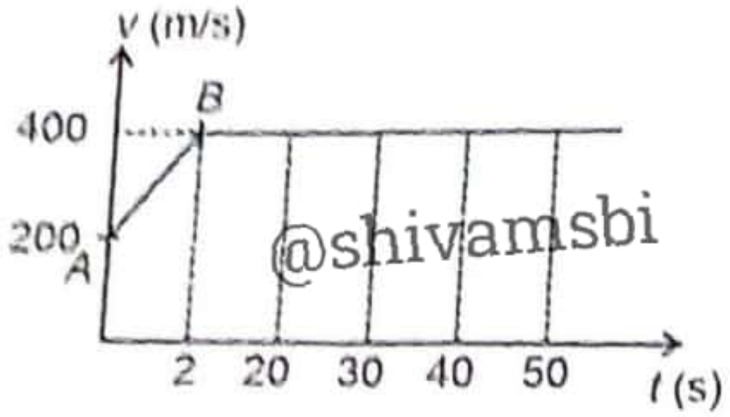
9. The position (x) of a particle moving on x-axis is given by relation, $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is
   (1) [L] (2) $[L^2]$
   (3) $[L^2T^{-2}]$ (4) $[L^3T^{-2}]$

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10. The motion of a body is represented by speed-time (v-t) graph as shown below:



The distance covered by body in first 30.5 seconds is (in km)

(1) 3 (2) 6

(3) 9 (4) 12

11. For an experiment, measured value of a quantity x is given by relation, $x = \frac{32.3 \times 1125}{27.4}$ where all digits are significant. If we want to report the value of x, we should write x as

(1) $x = 1356.19$

(2) $x = 1330$

(3) $x = 1326.186$

(4) $x = 1326.2$

12. The least count of a screw gauge is 0.01 mm. If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, then the new least count of screw gauge will be

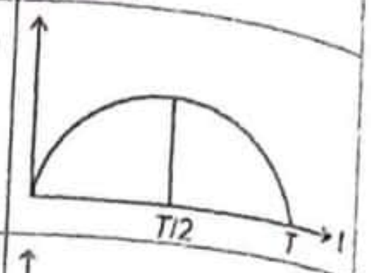
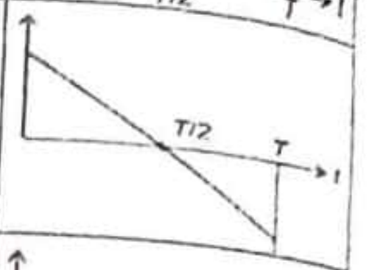
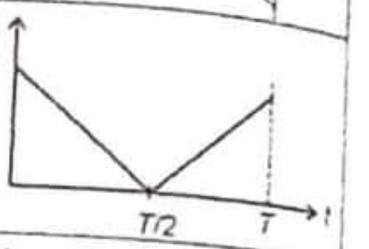
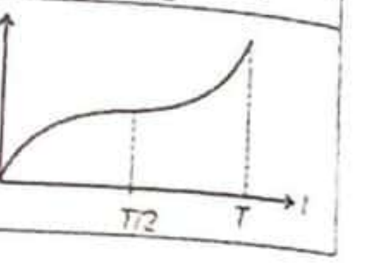
(1) 0.015 mm

(2) 0.035 mm

(3) 0.025 mm

(4) 0.045 mm

13. A body is thrown vertically up with speed u from the ground. Based on above information some graphs are mentioned in list-I and their plots are in list-II. If T denotes time of flight, then match the entries in list-I with entries in list-II.

	List-I		List-II
A.	Velocity-time graph	(P)	
B.	Speed-time graph	(Q)	
C.	Displacement-time graph	(R)	
D.	Distance-time graph	(S)	

Choose the **correct** option.

(1) A(Q), B(R), C(P), D(S)

(2) A(Q), B(P), C(R), D(S)

(3) A(R), B(Q), C(S), D(R)

(4) A(P), B(S), C(Q), D(R)

14. A particle is moving along x-axis such that its position(x) at time (t) is given by, $\sqrt{x} = 2t + 5$. The nature of motion of particle will be

(1) Uniformly accelerated motion

(2) Uniform motion

(3) Variable accelerated motion

(4) At rest

15. Consider a particle moving with uniform velocity and based on that read the following statements:

Statement (A): Average speed = Magnitude of average velocity

Statement (B): Instantaneous speed = Magnitude of instantaneous velocity

Statement (C): Distance covered = Magnitude of displacement

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In light of above statements, choose the correct option.

(1) All statements are correct

(2) Only statement A and B are correct

(3) Only statement B and C are correct

(4) Only statement A and C are correct

16. A particle starts moving from rest with constant acceleration and it covers a distance S_1 in the third second and distance S_2 in fifth second. The ratio $\frac{S_1}{S_2}$ is

(1) $\frac{3}{5}$ (2) $\frac{5}{9}$

(3) $\frac{9}{25}$ (4) $\frac{25}{81}$

17. A body traversed one-third of the distance with speed v_0. The remaining part of the distance was covered with speed v_1 for half the time and with speed v_2 for the other half of the time. The average speed of the body averaged over whole time of motion is

(1) $\frac{v_0 + v_1 + v_2}{3}$ (2) $\frac{2v_0 + v_1 + v_2}{3}$

(3) $\frac{3v_0(v_1 + v_2)}{v_1 + v_2 + 4v_0}$ (4) $\frac{3v_0(v_1 + v_2)}{v_1 + v_2 + 2v_0}$

18. A particle moving in a straight line with uniform acceleration finally comes to rest. What is the angle between its initial velocity and acceleration?

(1) Zero (2) $\frac{\pi}{4}$ radian

(3) $\frac{\pi}{2}$ radian (4) π radian

19. A man runs at a speed of 4 m s^{-1} to catch a bus. When he is 10 m behind the rear door, the bus begins to move with a constant acceleration of 1.2 m s^{-2}. The time after which man will catch the bus is

(1) 3.28 s

(2) 2.83 s

(3) 4.72 s

(4) He will never be able to catch the bus

20. The time elapsed is plotted along x-axis and the acceleration is plotted along y-axis. The area bounded between the graph and x-axis gives

(1) Average velocity

(2) Distance covered

(3) Change in velocity

(4) Change in acceleration

21. When the speed of car is v, the minimum distance over which it can be stopped is x. If the speed becomes nv, then the minimum distance over which it can be stopped is (Assume braking retardation to be same in both cases)

(1) $\frac{x}{n}$ (2) nx

(3) $\frac{x}{n^2}$ (4) n^2x

22. A particle has an initial velocity 9 m/s due east and a constant acceleration 2 m/s^2 due west. The distance covered by the particle in the fifth second of its motion is

(1) Zero (2) 0.5 m

(3) 2 m (4) 1 m

23. Let v and a respectively denote the velocity and acceleration of a body at any instant of time, then we can say that

(1) a may be zero when $v = 0$

(2) a must be zero when $v = 0$

(3) a may be zero when $v \neq 0$

(4) Both (1) and (3)

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24. Consider the following statements.

Statement (A): When a moving body varies its speed continuously, its velocity must be variable and it must have some acceleration.

Statement (B): When a moving body varies its velocity continuously, its speed must be variable and it must have some acceleration.

Statement (C): When a moving body varies its velocity continuously, its speed may or may not vary but it must have some acceleration.

Based on above information, consider the following options and choose the correct one.

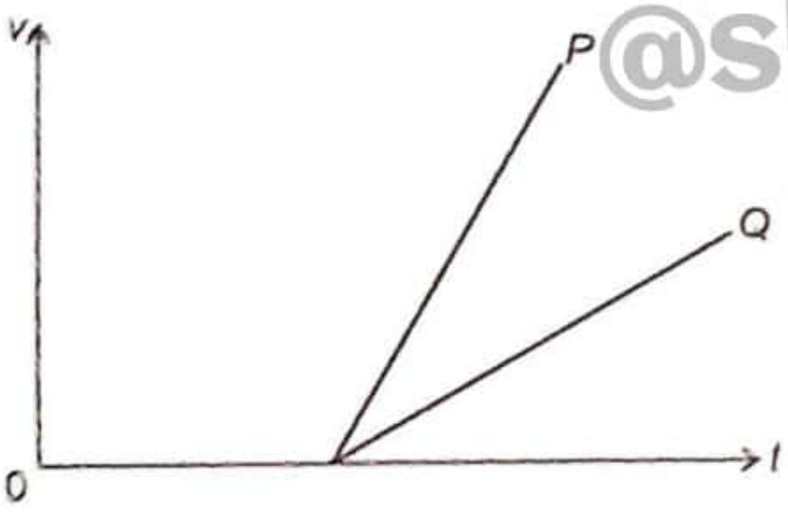
(1) All statements A, B and C are correct

(2) Only statements A and B are correct

(3) Only statements A and C are correct

(4) Only statement B is correct

25. Velocity-time (v-t) graph of the two particles P and Q moving in same direction are shown in the figure.



Based on above information, which of the following option about their magnitude of relative velocity is true?

(1) Relative velocity is zero

(2) Relative velocity is non-zero and constant

(3) Relative velocity is continuously increasing

(4) Relative velocity is continuously decreasing

26. The initial velocity of a particle is u at $t = 0$, and acceleration (A) is given by relation, $A = at$. Which of the following relation among given options is correct for the velocity (v) of particle at any time (t)? (where a is constant)

(1) $v = u + at$

(2) $v = u + at^2$

(3) $v = u + \frac{at^2}{2}$

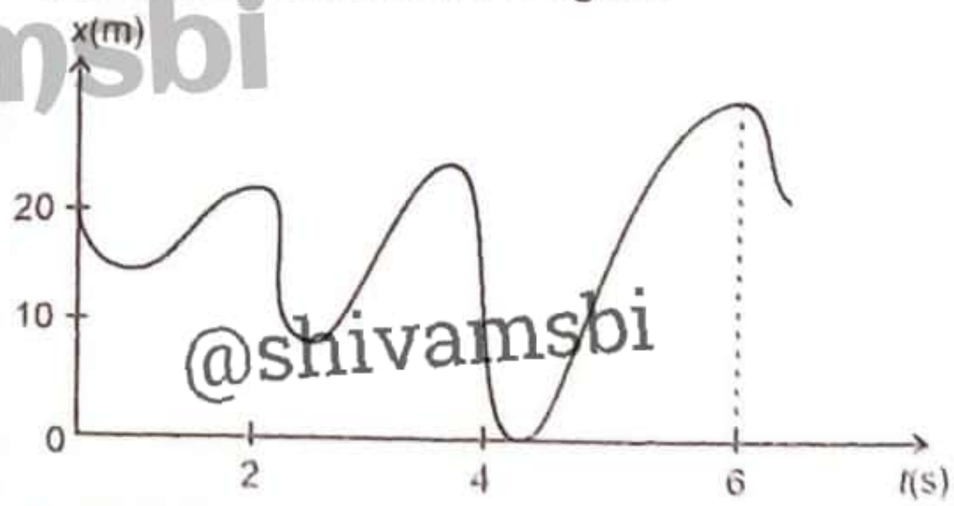
(4) $v = u + \frac{at^3}{3}$

27. Two balls are thrown upwards (from the ground) along the same vertical line at an interval of 5 seconds, with same initial velocity 40 m/s. These balls will collide at height of ($g = 10\ m/s^2$)

(1) 75 m (2) 60 m

(3) 48.75 m (4) 40 m

28. The position of a particle moving along x-axis varies with time as shown in the figure.



From information mentioned above, we can conclude that

(1) The particle has come to rest momentarily 6 times

(2) The maximum speed is at $t = 6$ second

(3) The velocity remains positive for $t = 0$ to $t = 6$ second

(4) The velocity remains negative for $t = 0$ to $t = 6$ second

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29. Three hot air balloons B_1, B_2 and B_3 are ascending with velocities v, $2v$ and $3v$ respectively. If a stone is dropped from each balloon when they are at same height, then

(1) Stone from B_1 reaches ground first

(2) Stone from B_2 reaches ground first

(3) Stone from B_3 reaches ground first

(4) The stones reach the ground simultaneously

30. If a car at rest accelerates uniformly to a speed 144 km/h in 20 second, then distance covered by car to attain this speed will be

(1) 20 m (2) 400 m

(3) 1440 m (4) 2880 m

31. Read the following two statements where one is labelled as **assertion** and another as **reason**.

Assertion (A): For a moving body, slope of distance-time graph is always positive.

Reason (R): Distance covered is always positive and increasing for a moving body.

Choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

32. Two trains A and B 100 m and 60 m long are moving in opposite directions on parallel tracks. The velocity of shorter train is 3 times that of the longer train. If trains take 4 seconds to cross each other, then velocities of trains are

(1) $v_A = 10$ m/s, $v_B = 30$ m/s

(2) $v_A = 2.5$ m/s, $v_B = 7.5$ m/s

(3) $v_A = 20$ m/s, $v_B = 60$ m/s

(4) $v_A = 5$ m/s, $v_B = 15$ m/s

33. Consider the equation for uniformly accelerated motion given as $s = ut + \frac{1}{2}at^2$.

Based on above information, consider following options and pick the correct one.

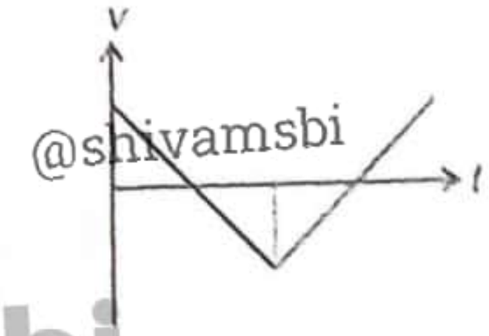
(1) s denotes the distance covered

(2) s denotes the displacement covered

(3) u is initial velocity at time $t = 0$, and is always positive

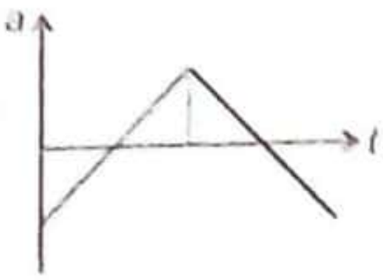
(4) a denotes constant acceleration and is always positive

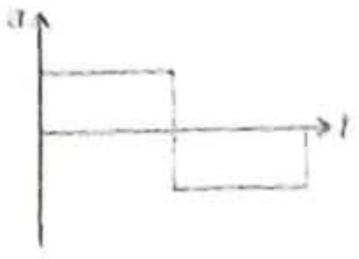
34. The figure given below shows the velocity (v) versus time (t) graph for a moving body.

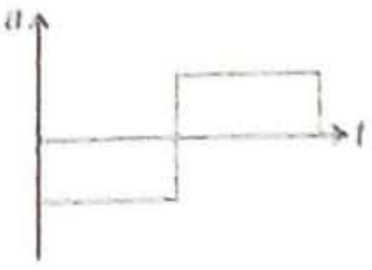


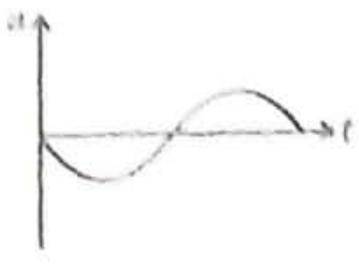
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Which of the following graph represents corresponding acceleration (a) versus time (t) graph?

(1) 

(2) 

(3) 

(4) 

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35. A body is projected vertically upwards from the ground. If t_1 and t_2 be the time at which body is at same height h above the point of projection while ascending and descending respectively, then height h is equal to

(1) $\frac{1}{2}gt_1t_2$ (2) gt_1t_2

(3) $g\frac{(t_1+t_2)}{2}$ (4) $g(t_1+t_2)$

36. A stone is thrown vertically upwards from the top of tower with initial speed u. It reaches the ground with speed $3u$. The time of flight of the stone is

(1) $\frac{2u}{g}$ (2) $\frac{4u}{g}$

(3) $\frac{3u}{g}$ (4) $\frac{u}{g}$

37. The position (x) of particle varies with time t as $x=\frac{1}{2}t^2-\frac{1}{3}t^3$, where t is in second. The acceleration of the particle will be zero at time t equal to

(1) 2 second (2) $\frac{1}{2}$ second

(3) $\frac{3}{2}$ second (4) $\frac{2}{3}$ second

38. A train accelerates from rest at constant rate α and covers distance s_1 in time t_1. After that it retards at constant rate β (and comes to rest) covering distance s_2 in time interval t_2.

Based on above information, which of the following relation is correct?

(1) $\frac{s_1}{s_2}=\frac{\alpha}{\beta}=\frac{t_1}{t_2}$ (2) $\frac{s_1}{s_2}=\frac{\beta}{\alpha}=\frac{t_1}{t_2}$

(3) $\frac{s_1}{s_2}=\frac{\alpha}{\beta}=\frac{t_2}{t_1}$ (4) $\frac{s_1}{s_2}=\frac{\beta}{\alpha}=\frac{t_2}{t_1}$

39. Amount of solar energy received on the Earth's surface per unit area per unit time is defined as solar constant. Dimension of solar constant is

(1) $[ML^2T^{-2}]$ (2) $[MLT^{-2}]$

(3) $[M^2L^0T^{-1}]$ (4) $[ML^0T^{-3}]$

40. The least count for stop watch is $\frac{1}{5}$ second. The time for 20 oscillations of a pendulum is measured as 15 second. The percentage error in the measurement of time period of pendulum will be

(1) 1.5% (2) 6.7%

(3) 1.3% (4) 2.3%

41. Given below are two statements: one of them is labelled as **assertion** and another is labelled as **reason**.

Assertion (A): Number of significant digits in 0.01 is one and that in 0.50 is two.

Reason (R): If number is less than 1, then zeroes before first non-zero digit are not significant and trailing zeroes are significant.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

42. The numbers 3.865 and 3.875 after rounding off to 2 significant figures will be reported as

(1) 3.9 (2) 3.8

(3) 3.7 (4) 4.0

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43. A physical quantity y is related to x by the relation, $y = k \tan x$; (where k denotes a positive constant). The fractional error in y is equal to

(1) $\frac{dx}{\sin 2x}$ (2) $\frac{2dx}{\sin 2x}$

(3) $\frac{dx}{\tan x}$ (4) $\frac{dx}{2\tan x}$

44. Which of the following operation on physical quantities A and B, will be meaningful, if A and B are both scalar and have different dimensions?

(1) $\frac{A}{B}$ (2) $\frac{A}{e^{(A-B)}}$

(3) $A - B$ (4) Both (1) and (2)

45. If the speed of light (c), acceleration due to gravity (g) and pressure (p) are taken as fundamental quantities, then dimensional formulae of gravitational constant (G) is

(1) $[c^0 g p^{-3}]$ (2) $[c^2 g^3 p^{-2}]$

(3) $[c^0 g^2 p^{-1}]$ (4) $[c^2 g^2 p^{-2}]$

[CHEMISTRY]

46. Consider the following mixtures

a. aq. NaCl

b. Air

c. Milk

d. Iron filings

Homogeneous mixtures are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c) only

(4) (a), (c) and (d) only

47. Match the units in list I with base physical quantity in list II.

	List I		List II
a.	Mole	(i)	Mass
b.	Kilogram	(ii)	Luminous intensity
c.	Candela	(iii)	Amount of substance
d.	Ampere	(iv)	Electric current

Correct match is

(1) a(iii), b(i), c(ii), d(iv) (2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(iv), d(ii) (4) a(i), b(iii), c(iv), d(ii)

48. Number of molecules of CO left if 2.1 mg CO is removed from 2.24 mL of CO gas at STP will be

(1) 1.5×10^{22} (2) 4.5×10^{19}

(3) 4.5×10^{22} (4) 1.5×10^{19}

49. 1.8062 g of a mixture of NaCl and KCl on treatment with H_2SO_4 gave 2.1568 g of a mixture of Na_2SO_4 and K_2SO_4. Approximate percentage composition of NaCl and KCl respectively is

(1) 42.64% and 57.36%

(2) 57.36% and 42.64%

(3) 22.64% and 77.36%

(4) 77.36% and 22.64%

50. A crystalline salt $Na_2SO_4 \cdot xH_2O$ on heating loses 55.9% of it's mass and becomes anhydrous. The formula of crystalline salt is

(1) $Na_2SO_4 \cdot 10H_2O$ (2) $Na_2SO_4 \cdot 2H_2O$

(3) $Na_2SO_4 \cdot 7H_2O$ (4) $Na_2SO_4 \cdot 5H_2O$

51. A mixture of 4.6 g formic acid and 9 g oxalic acid is treated with conc. H_2SO_4. The evolved gaseous mixture is passed through KOH pellets. Weight (in gram) of the remaining gaseous product at STP will be

(1) 2.8 g (2) 10 g

(3) 5.6 g (4) 11.2 g

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52. If 2 g H_2 is mixed with 14 g N_2 to produce NH_3 in a closed vessel.

Consider the following statements

(a) Mass of reaction mixture will be 16.0 g at the end of the reaction.

(b) 16 g NH_3 will be produced.

(c) Volume of NH_3 evolved at STP will be nearly 21.08 L.

(d) H_2 is limiting reagent.

Correct statements are

(1) (a), (b) and (c) only

(2) (a) and (d) only

(3) (a), (c) and (d) only

(4) (c) and (d) only

53. Given below are two statements

Statement I: Compounds with same empirical formula have similar percentage composition of elements.

Statement II: Empirical formula of benzene and acetylene are same.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

54. If the molality of an aqueous solution of urea is 3 m then the mass of water present in 1.8 kg of this solution will be approximately

(1) 1.2 kg (2) 1.1 kg

(3) 1.0 kg (4) 1.5 kg

55. Consider the following statements

(a) Matter is made up of extremely small indivisible particles called atoms.

(b) All atoms of a given element have identical mass.

(c) Atom can neither be created nor be destroyed in a chemical reaction.

(d) Atoms of different element combine with each other in a variable whole number ratio to form compounds.

Correct statements according to Dalton's atomic theory are

(1) (a), (b) and (d) only (2) (a) and (c) only

(3) (a), (b) and (c) only (4) (b) and (d) only

56. How many grams of concentrated HNO_3 solution should be used to prepare 700 mL of 3.0 M HNO_3 solution?

[Concentrated nitric acid is 70% (w/w) HNO_3]

(1) 364 g (2) 274 g

(3) 189 g (4) 145 g

57. Consider the following statements

a. Both 6.00 and 600 have three significant figures.

b. H_2O and H_2O_2 follow law of multiple proportions.

c. Mass percentage of oxygen in glucose molecule is 45%.

Correct statement(s) is/are

(1) (c) only (2) (a) and (b) only

(3) (b) only (4) (a) and (c) only

58. Density in g/L of N_2 gas at STP will be

(1) 2.5 (2) 0.625

(3) 1.25 (4) 1.87

59. Number of mole of H^+ ions present in a mixture prepared by mixing 200 mL of 0.2 M H_2SO_4, 300 mL of 0.1 M HNO_3 and 500 mL of 0.4 M HCl is

(1) 0.15 (2) 0.93

(3) 0.62 (4) 0.31

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60. Consider the following statements for 0.15 g of CO_3^{2-} ions

a. Number of ions are nearly 1.5×10^{22}.

b. Number of O atoms are 4.5×10^{23}.

c. Mass of carbon is 0.03 g.

Correct statement(s) is/are

(1) (a) and (b) only (2) (a) and (c) only

(3) (c) only (4) (a), (b) and (c) only

61. Minimum molecular mass of a macromolecule that contains 0.01% of Mg by mass is

(1) 240 kg (2) 10^4 u

(3) 10 kg (4) 2.4×10^5 u

62. A compound contains 54.54% carbon, 36.37% oxygen and rest hydrogen. Empirical formula mass of compound will be

(1) 88 u (2) 44 u

(3) 66 u (4) 30 u

63. For complete combustion of 3 g ethane, mass of O_2 required and volume of CO_2 obtained at STP respectively will be

(1) 11.2 g and 4.48 L (2) 16 g and 4.48 L

(3) 11.2 g and 2.24 L (4) 16 g and 2.24 L

64. 4 g of each Mg and S are taken to produce MgS. Mass of MgS produced will be

(1) 1 g (2) 8 g

(3) 6 g (4) 7 g

65. Match the temperature in °C in List I with °F in List II.

	List I		List II
a.	25°C	(i)	104°F
b.	40°C	(ii)	77°F
c.	–60°C	(iii)	212°F
d.	100°C	(iv)	–76°F

Choose the **correct** option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iii), b(i), c(iv), d(ii)

66. Equivalent weight of a metal oxide that contains 40% (w/w) oxygen is

(1) 24 (2) 12

(3) 20 (4) 40

67. 100 mL decinormal H_2SO_4 is mixed with 100 mL seminormal HCl, volume of 0.2 M NaOH solution required to neutralise mixture obtained will be

(1) 600 mL (2) 300 mL

(3) 100 mL (4) 200 mL

68. Consider the following statements

a. According to JJ Thomson, atom is a sphere of uniform positive charge and all the negative charge is clustered at the centre of atom.

b. Rutherford's α-particle scattering experiment proved that atom has a nucleus.

c. During α-particle scattering experiment most of the particles got deflected.

d. Ratio of radius of nucleus to radius of atom is nearly 10^{-5}.

e. Mass of proton is nearly 1837 times the mass of electron.

Correct statements are

(1) (a) and (c) only

(2) (c), (d) and (e) only

(3) (b) and (e) only

(4) (b), (d) and (e) only

69. For two isobars 'A' and 'B' with atomic number 48 and 52 respectively, if mass number of A is 100 then number of neutrons in element B will be

(1) 48 (2) 78

(3) 50 (4) 52

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70. An element Y exists in two isotopic forms ^{20}Y and ^{22}Y. If average atomic mass of element is 20.3 amu then percentage abundance of heavier isotope will be

(1) 85% (2) 70%

(3) 30% (4) 15%

71. Match element in list I with quantum number of last filled electron in list II.

	List I		List II
a.	Mg	(i)	$n = 4, l = 0, m = 0, s = +\frac{1}{2}$
b.	K	(ii)	$n = 3, l = 2, m = +1, s = +\frac{1}{2}$
c.	Br	(iii)	$n = 3, l = 0, m = 0, s = +\frac{1}{2}$
d.	Mn	(iv)	$n = 4, l = 1, m = -1, s = +\frac{1}{2}$

Choose the correct option.

(1) a(i), b(iii), c(iv), d(ii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iv), b(iii), c(ii), d(i)

72. If an electron has equal value of uncertainty in position and uncertainty in velocity then minimum possible uncertainty in its momentum will be

(1) $\sqrt{\frac{h}{4\pi m}}$ (2) $\sqrt{\frac{hm}{4\pi}}$

(3) $\sqrt{\frac{4\pi}{hm}}$ (4) $\sqrt{\frac{4\pi m}{h}}$

73. de-Broglie wavelength of an electron with kinetic energy of 2.8×10^{-23} J will be approximately

(1) 0.93×10^{-10} m (2) 0.93×10^{-7} m

(3) 0.93×10^{-21} m (4) 0.93×10^{-8} m

74. Which among the following set of quantum numbers are possible?

(a) $n = 4, l = 1, m = -1, s = -1$

(b) $n = 2, l = 1, m = +2, s = \frac{-1}{2}$

(c) $n = 5, l = 4, m = -3, s = +\frac{1}{2}$

(d) $n = 3, l = 2, m = +2, s = -\frac{1}{2}$

(e) $n = 2, l = 1, m = +2, s = +\frac{1}{2}$

Choose the correct option.

(1) (a), (b), (d) and (e) only

(2) (b), (d) and (e) only

(3) (a), (b) and (d) only

(4) (c) and (d) only

75. Given below are two statements

Statement I: d_{xy} orbital has two angular nodes.

Statement II: In 5*p* orbital total number of nodes are 4.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

76. Consider the following statements

a. Number of degenerate orbitals in 2nd excited state of H-atom is 9.

b. Energy of 2*s* orbital for Na atom is lesser than energy of 2*s* orbital of Li atom.

c. Order of energy of orbitals for Be^{3+} ion is $3d > 4s > 3p > 3s$

d. Splitting of spectral line in an electric field is known as Stark effect.

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Correct statements are

(1) (a), (b) and (d) only (2) (a) and (b) only

(3) (c) and (d) only (4) (b) and (d) only

77. The maximum number of electrons present with $m_l = 0$ in Cu atom is

(1) 10 (2) 11

(3) 12 (4) 13

78. The correct number of unpaired electrons present in the element with atomic number 24 is

(1) 4 (2) 5

(3) 6 (4) 7

79. Which among the following is not correctly matched according to the property indicated?

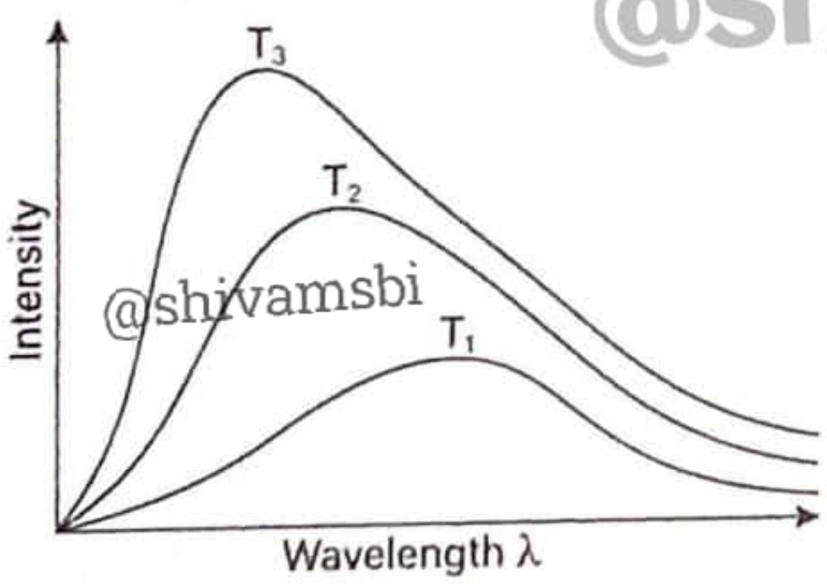
(1) Electron > Proton > Neutron : Magnitude of charge/mass ratio

(2) γ-rays > β-rays > α-rays : Penetrating power

(3) Infrared > Radiowave > Microwave : Frequency

(4) $H > He^{+} > Li^{2+}$: Radius of first orbit

80. The graph shows the variation of intensity with wavelength at different temperatures for a perfect black body



The correct decreasing order of temperature is

(1) $T_1 > T_2 > T_3$ (2) $T_3 > T_2 > T_1$

(3) $T_2 > T_1 > T_3$ (4) $T_2 > T_3 > T_1$

81. Kinetic energy of photoelectron ejected by a photon with 10^{15} s^{-1} frequency from a metal with work function of 3.44×10^{-19} J is

(1) 5.4×10^{-19} J (2) 3.2×10^{-19} J

(3) 6.8×10^{-18} J (4) 1.4×10^{-18} J

82. Energy of 2nd excited state of Li^{2+} ion is x J then energy of 3rd energy level of Be^{3+} ion will be

(1) $\frac{9x}{16}$ J (2) x J

(3) $\frac{x}{3}$ J (4) $\frac{16x}{9}$ J

83. Given below are two statements one is assertion (A) and other is reason (R).

Assertion (A): Principal quantum number of 2nd excited state of H^- ion is n = 3.

Reason (R): Energy of orbitals for H atom depends only on principal quantum number.

In the light of above two statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

84. Wave number of a photon emitted due to transition of electron from 5th orbit to second orbit in He^+ ion is

(1) $R_H \frac{21}{25}$ (2) $R_H \frac{16}{25}$

(3) $R_H \frac{25}{21}$ (4) $R_H \frac{25}{16}$

85. Number of carbon atoms present in 200 mL of 0.25 M aqueous solution of glucose is

(1) 1.2 N_A (2) 0.8 N_A

(3) 0.5 N_A (4) 0.3 N_A

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86. Mass of $BaCl_2$ required to produce 23.3 g barium sulphate on reaction with 18.2 g sodium sulphate will be

(Atomic mass of Ba = 137 u)

(1) 20.8 g (2) 10.4 g

(3) 41.6 g (4) 5.7 g

87. If a photon of 11 eV is supplied to H-atom then in which energy level electron will be present?

(Considering electron is present in ground state of H-atom)

(1) 2nd (2) 3rd

(3) 1st (4) 4th

88. For a hydrogen atom, if an electron is present in 5th excited state, then the number of visible line(s) present in emission spectrum is/are

(1) 1 (2) 5

(3) 6 (4) 2

89. de-Broglie wavelength of an electron in 2nd excited state of Li^{2+} ion will be

(1) 1.058π Å (2) 0.529π Å

(3) 2.116π Å (4) 0.26π Å

90. Which among the following transition will emit photon with maximum frequency for Li^{2+} ion?

(1) $\infty \rightarrow 2$ (2) $2 \rightarrow 1$

(3) $8 \rightarrow 3$ (4) $3 \rightarrow 1$

[BOTANY]

91. [redacted] is responsible for

(1) Inheritance of characters and intracellular digestion of essential biomolecules

(2) ATP and sugar synthesis

(3) Active ribosomal RNA synthesis

(4) Origin of plasmodesmata and desmotubules

92. Which among the following sequences **correctly** represents the pathway followed by newly synthesised protein molecule in a eukaryotic cell?

(1) RER → Transitional vesicles → Golgi apparatus → Secretory vesicles → Plasma membrane.

(2) Ribosome → Golgi apparatus → SER → Plasma membrane → Secretory vesicles.

(3) RER → Golgi apparatus → SER → Plasma membrane → Secretory vesicles.

(4) SER → Secretory vesicles → Plasma membrane → Golgi apparatus.

93. A prokaryotic cell exhibits the following features:

a. Multiple ribosomes are translating a single transcript.

b. Photosynthetic pigments are embedded in the internal membrane folds.

c. Small bristle like fibres that help the cell adhere strongly to a host tissue.

Which among the following combinations of structures best explains the observations 'a', 'b' and 'c' respectively?

(1) Polysome, mesosome and pili

(2) Palade particles, chromatophore and cilia

(3) Nucleosome, thylakoids and fimbriae

(4) Polyribosome, chromatophore and fimbriae

94. Select the **correctly** matched pair from the following.

(1) Chloroplast – [redacted] of genetic material

(2) Centrosome – Usually contains two pairs of cylindrical structures

(3) Cytoskeleton – Helps in motility in [redacted]

(4) Chromatin – Contains DNA and some acidic proteins, like histones

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95. A single membrane bound organelle in eukaryotic cell is

(1) Chloroplast (2) Lysosome

(3) Nucleus (4) Mitochondrion

96. Read the following Assertion (A) and Reason (R) statements and choose the **correct** option.

Assertion (A): Core of eukaryotic flagella and cilia has nine doublets of radially arranged peripheral microtubules.

Reason (R): The basal body of eukaryotic cilia or flagella has 9+2 organisation of microtubules.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

97. Match the column I with column II and select the **correct** option.

	Column I		Column II
A.	Vacuole	(i)	It is responsible for trapping light energy essential for photosynthesis.
B.	Mitochondria	(ii)	It helps in the assembly of amino acid into polypeptide chains.
C.	Chloroplast	(iii)	It contains water, sap, excretory products and other materials not useful for cell.
D.	Ribosome	(iv)	The matrix of this organelle contains single circular DNA.

	A	B	C	D
(1)	ii	i	iv	iii
(2)	iii	iv	i	ii
(3)	iii	i	ii	iv
(4)	iv	iii	i	ii

98. Select the **incorrect** match w.r.t plasma membrane of eukaryotes.

(1) Integral proteins - Partially or totally buried in lipid bilayer

(2) Non-polar tail of lipids - Projected towards the aqueous environment.

(3) Phospholipids - Primary constituent of lipid component

(4) Peripheral proteins - Can be easily extracted from a membrane.

99. A living denucleated eukaryotic cell, still may have some genetic material. This genetic material would be in

(a) Plasmid

(b) Nucleoid

(c) Chloroplast

(d) Peroxisome

(e) Mitochondria

The **correct** ones are

(1) (a), (c) and (e) (2) (c) and (e)

(3) (b) and (d) (4) (a), (d) and (e)

100. Which of the following statements about [redacted] is **correct**?

(1) Principally performs the function of packaging of material.

(2) It is important for osmoregulation and excretion like in *Amoeba*.

(3) Helps in the formation of cell wall and DNA replication.

(4) It is involved in the synthesis of cholesterol.

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101. Which among the following layer of cell envelope in bacteria is responsible for giving them [redacted]?
(1) Slime layer (2) Plasma membrane
(3) Cell wall (4) Capsule

102. Consider the following statements of Assertion and Reason.

Assertion (A): Cell cycle is a sequence of events by which a cell duplicates its genome, synthesizes the other constituents of the cell and eventually divides into two daughter cells.

Reason (R): The replicated chromosomes are distributed to daughter nuclei by a complex series of events during cell division.

In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true statement but (R) is false
(4) Both (A) and (R) are false statements

103. [redacted] is marked by
(1) Formation of synapse of homologous chromosome.
(2) Appearance of X shaped structure called chiasmata.
(3) Gradual decondensation of chromosome.
(4) Terminalisation of chiasmata

104. Which among the following is **true** for the organelle called "Power house of the cell"?
(1) The folds of the inner membrane increase the surface area for oxidative phosphorylation.
(2) Protein synthesis within this organelle is carried out exclusively by cytosolic ribosomes.
(3) The matrix contains linear DNA associated with histone proteins.
(4) Enzymes of the electron transport system are localised predominantly within the matrix.

105. The organelle that is **not** included in the endomembrane system is
(1) Vacuole
(2) Lysosome
(3) Endoplasmic Reticulum
(4) Peroxisome

106. Which among the following pigments is [redacted] and found in the plant cell vacuole?
(1) Xanthophylls
(2) Carotenoids
(3) Anthocyanin
(4) Chlorophyll

107. Select the **correctly** matched pair.
(1) Presence of carotenoid pigments [redacted] of DNA and RNA – Chloroplast
(2) Can be removed by detergents – Integral membrane protein
(3) Site of photosynthesis in [redacted] – Chromatophore
(4) Layer deposited external to the primary cell wall – Secondary cell wall

108. The type of chromosome in which the centromere is present very close to one end of the chromosome is called
(1) Sub-metacentric
(2) Acrocentric
(3) Metacentric
(4) Telocentric

109. In a cell, the bivalent has
(1) Four chromatids of the two homologous chromosomes
(2) Two pairs of homologous chromosomes
(3) Four centromere of the four chromosomes
(4) Two chromatids of the two non-homologous chromosomes

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110. The [redacted] differs from the [redacted] as in the former,
(1) There is half the number of chromosomes as compared to the latter one
(2) Bivalent chromosomes get aligned on the equatorial plate
(3) Each chromosome has two chromatids
(4) Microtubules are attached to the centromere of the chromsomes

111. The cell cycle is divided into how many basic phases?
(1) Four (2) Three
(3) Two (4) Five

112. Match the scientists in Column I with their scientific contributions in Column II.

	Column I		Column II
A.	Robert Brown	I.	Proposed that every cell arises from a pre-existing cell
B.	[redacted]	II.	Identified the nucleus
C.	Camillo Golgi	III.	Concluded that plants are composed of different kinds of cells
D.	[redacted]	IV.	He identified densely stained reticular structures in the cytoplasm near the nucleus.

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Choose the **correct** combination.
(1) A–II, B–III, C–IV, D–I
(2) A–III, B–II, C–IV, D–I
(3) A–II, B–I, C–III, D–IV
(4) A–IV, B–III, C–II, D–I

113. Which among the following is **true** for [redacted] stage of interphase of cell cycle?
(1) The number of chromosomes will be more as compared to G_2 phase.
(2) Replication of the nuclear genome has already begun.
(3) Most of the membrane bound organelle increase in number.
(4) Organelles like mitochondria and plastids get distributed between the two daughter cells.

114. Read the following statements of Assertion **(A)** and Reason **(R)** and choose the **correct** option.
Assertion (A): The successful completion of the mitosis in a eukaryotic cell does not necessarily require the presence of the centriole.
Reason (R): Spindle microtubules can assemble in the absence of the centrioles in several eukaryotic cells, including those of higher plants.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false statements
(4) (A) is true but (R) is false

115. Two dividing cells P and Q of organisms having complex body organisation are observed by biologist.
Cell P: Cytoplasm divides by inward constriction of the plasma membrane.
Cell Q: Cytoplasmic division is achieved due to the formation of phragmoplast by Golgi-derived vesicles.
Which conclusion is most appropriate for the above cells?
(1) P and Q show different phases of mitosis.
(2) Q possesses a rigid cell wall that maintains the shape of the cell.
(3) P lacks spindle fibres.
(4) Q cannot undergo telophase.

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116. A diploid meiocyte completes meiosis and produces four daughter cells. Each daughter cell contains [redacted] chromosomes with [redacted] DNA content. The cell formed immediately after the completion of meiosis I contained
(1) 46 chromosomes and 1C DNA
(2) 23 chromosomes and 2C DNA
(3) 46 chromosomes and 2C DNA
(4) 23 chromosomes and 4C DNA

117. Which among the following is [redacted] significance of [redacted] division?
(1) Cell repair
(2) Maintaining the nucleo-cytoplasmic ratio of the cell
(3) Increasing the genetic variability in the daughter cell
(4) Repairing of the upper layer of the epidermis, cells of the lining of gut and blood cells

118. [redacted] does **not** consist of
a. Spindle fibres
b. Basal body and hook
c. Centromere
d. Centrosome
e. Microtubules
The **correct** ones are
(1) (a) and (e) only (2) (b), (c) and (d)
(3) (b) and (c) only (4) (a), (d) and (e) only

119. What will be next major event during meiosis just after [redacted]?
(1) Separation of homologous chromosomes from each other
(2) Alignment of the centrosome at the cell equator through the attachment of spindle fibres
(3) Initiation of centriole duplication for the cell division
(4) Shortening of interzonal fibres and movement of sister chromatids towards opposite poles

120. Read the following statements stating them as **True (T)** or **False (F)** and choose the **correct** option.
A. M phase of the cell cycle involves nuclear division but not the cytoplasmic division.
B. In eukaryotic organisms, growth is considered in terms of number of cells but not in terms of cytoplasmic content.
C. In plants, mitotic division can occur in both haploid and diploid cells.

	A	B	C
(1)	F	T	T
(2)	T	T	F
(3)	F	F	T
(4)	F	F	F

121. Which among the following statements is **true** for [redacted]?
(1) Cytokinesis will occur, resulting in the formation of four daughter cells.
(2) Chromosomal DNA undergoes replication to prepare the cell for the second round of division.
(3) Synthesis of RNA and proteins occurs which is required for the subsequent meiotic division to continue.
(4) In animal cells, centrioles and chromosome duplicate before the onset of the second round of division.

122. The microtubules radiate out of the centrosomes are called
(1) Phragmoplasts (2) Interzonal fibres
(3) Asters (4) Kinetochores

123. The cells during [redacted] phase
(1) Are metabolically active and continuously grow but does not replicate their DNA
(2) Divide their cytoplasm into two equal proportion
(3) Divide in the random manner
(4) Duplicates its most of the cell organelle including Golgi apparatus

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124. The centromere of each chromosome split during A of mitosis and B of meiosis.

Choose the **correct** option to fill in the blanks A and B, respectively.

(1) Anaphase and Anaphase I
(2) Metaphase and Telophase
(3) Prophase and Telophase I
(4) Anaphase and Anaphase II

125. [redacted] involves

(1) Single cycle of nuclear division
(2) Single cycle of DNA replication
(3) Single cycle of cytoplasmic division
(4) Two cycles of DNA replication

126. [redacted] of a prokaryotic cell

(1) Is made up of polysaccharide
(2) Protects cell from bursting
(3) Determines the shape of the cell
(4) Is not responsible for giving gummy and sticky character to a cell

127. Which among the following structures of the cell connects the cytoplasm of one cell to other?

(1) Nuclear pore (2) Plasmodesmata
(3) Middle lamella (4) Plasma membrane

128. Select the function of an organelle that is exclusive to eukaryotic cell and it is not surrounded by any membrane.

(1) Formation of plasma membrane
(2) Glycosylation of proteins
(3) Formation of spindle fibres
(4) Synthesis of enzymatic proteins

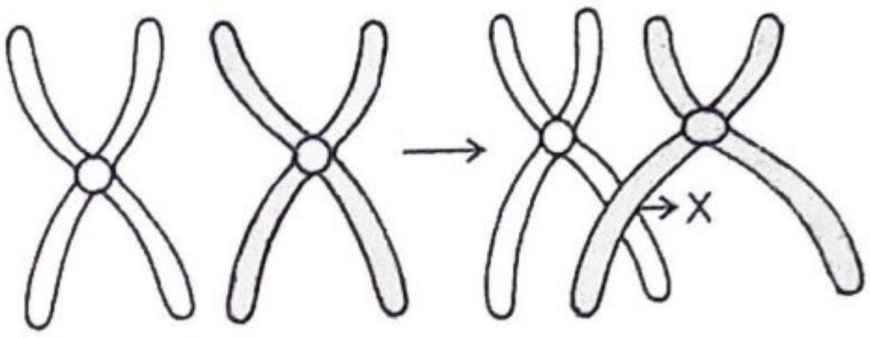
129. During which phase of the cell cycle, the cells remain metabolically active but do not proliferate unless they are called on to do so?

(1) G_1 phase (2) S phase
(3) G_2 phase (4) G_0 phase

130. [redacted] ends with the completion of

(1) Telophase
(2) Metaphase
(3) Cell plate formation
(4) Anaphase

131. In the given diagram, an event associated with chromosomes during meiosis is shown.



A. X is visible after the dissolution of synaptonemal complex.
B. The formation of region 'X' is preceded by exchange of genetic material between non-sister chromatids of the homologous chromosome.
C. The region 'X' is established only after homologous chromosomes completely separate during diakinesis.

Select the statement(s) which is/are **true** for the above structure.

(1) All A, B and C (2) A and C only
(3) A and B (4) B only

132. [redacted] vacuole in [redacted]

(1) Is important for excretion
(2) Provide buoyancy
(3) Is covered by double membrane
(4) Can store food material

133. The [redacted] subunit of ribosomes found in semi-autonomous cell organelles is

(1) 70S (2) 50S
(3) 40S (4) 80S

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134. Select the **correctly** matched pair.
(1) White Blood cells - Elongated
(2) Nerve cell - Unbranched and short.
(3) Mesophyll cell - Round and oval
(4) Red blood cell - Sickle shaped and branched

135. The flattened membranous sacs of chloroplast
(1) Have enzymes required for protein synthesis
(2) Are involved in the breakdown of the lipids
(3) Are concerned with the modification of proteins
(4) Contain chlorophyll pigments

[ZOOLOGY]

136. All of the following secretions of glands are released through ducts to their target site, **except**
(1) Bile juice (2) Sweat
(3) Hormone (4) Saliva

137. Complete the analogy by selecting the correct option.
Tight junction : Prevents leakage of substances across a tissue :: Gap junction : _____
(1) Cements neighbouring cells together to resist mechanical stress
(2) Anchoring cells to the basement membrane *via* protein filaments
(3) Enables communication between adjacent cells for transfer of ions and molecules
(4) Prevents entry of pathogen across the epithelial layer

138. A researcher studying the epithelial lining in humans noticed that a particular epithelium has the following features:
(a) Cells are tall and slender
(b) Nuclei are located at the base
(c) Free surface bears hair-like motile projections
The epithelium having above mentioned features is most likely found in the inner surface of structure that aids in
(1) Transport of gametes
(2) Digestion of food
(3) Excretion of nitrogenous wastes
(4) Exchange of gases

139. Which of the following correctly describes the functional logic behind the tight junctions found in the apical part of the adjacent intestinal epithelial cells?
(1) They are present here to allow rapid transport between enterocytes for coordinated peristalsis
(2) They seal the intercellular spaces to prevent passage of cytoplasmic contents.
(3) They are present here exclusively to anchor cells to the basement membrane
(4) They allow selective permeability of substances.

140. 'X' constitutes second largest percentage of the total cellular mass. All of the following belong to the category of 'X', **except**
(1) Trypsin
(2) Antibody
(3) Inulin
(4) Collagen

141. In a polysaccharide chain like cellulose,
(1) The right end is reducing end due to the presence of free aldehyde group
(2) The left end is non-reducing end due to the presence of free ketone group
(3) The right end is called the C-terminal end
(4) The left end is called the N-terminal end

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142. In a DNA strand, a purine base pairs with a pyrimidine base by double H-bonds. The pyrimidine in the question is 'P'. Instead of 'P', _____ is present in a RNA strand.

Select the option that correctly fills the blank.

(1) Guanine (2) Thymine

(3) Cytosine (4) Uracil

143. Choose the secondary metabolite among the following which is a polymeric substance and is present in the plant cell wall.

(1) Monoterpene (2) Cellulose

(3) Vinblastin (4) Concanavalin A

144. Read the given statements and select the correct option.

Statement (A): Both tendons and ligaments are classified as dense regular connective tissues, differing only in the structures they connect.

Statement (B): In areolar tissue, fibres are compactly and regularly arranged, making it a connective tissue with minimal ground substance.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

145. Choose the incorrect statement w.r.t. muscle fibres in the human body.

(1) Cardiac muscle fibres are branched, cylindrical, uninucleated and show striations.

(2) Smooth muscle fibres are spindle-shaped, involuntary and lack striations, and they are well-suited for slow and sustained contractions.

(3) Skeletal muscle fibres are multinucleated, striated and voluntary, with nuclei present at the periphery.

(4) All the three types of muscle fibres possess intercalated discs.

146. A medical student is examining a histological slide. She observes the following features:

(a) Fibres of this tissue are loosely intertwined between cells

(b) Contains several types of cells including plasma cells, mast cells, macrophages, eosinophils

(c) Ground substance is semi-fluid

The tissue she is examining is involved in

(1) Providing structural support to the organs

(2) Forming a protective lining over internal and external surfaces of the body

(3) Generating electrical impulses in response to a stimuli

(4) Playing an active role in all the primary movements of the body

147. Match Column I with Column II and select the correct option.

	Column I		Column II
a.	Adipose tissue	(i)	Lacks its own blood supply and exhibits a high rate of cell division
b.	Neural tissue	(ii)	Maintains the positions of the various parts of the body
c.	Epithelial tissue	(iii)	Exerts the greatest control over the body's responsiveness to changing conditions
d.	Muscular tissue	(iv)	Reduces heat loss *via* skin and serves as an energy reserve

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(i), b(iii), c(ii), d(iv)

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148. All of the following are correct w.r.t. cartilage, **except**

(1) The ground substance is solid and pliable due to deposition of chondroitin sulphate

(2) Can endure considerably more compressive strain than loose and dense connective tissues.

(3) Chondrocytes occur singly or in groups within spaces called lacunae in the matrix

(4) Some of them are responsible for haematopoiesis *i.e.*, production of blood cells

149. Consider the given statements and select the correct option.

Statement (A): The percentage weight of carbon in Earth's crust is exactly equal to the percentage weight of sulphur in Earth's crust.

Statement (B): Dry weight of a small living tissue is obtained by fully burning the tissue, as all the carbon compounds are oxidised to gaseous form [CO_2, water vapour] and are removed.

(1) Only statement (A) is correct

(2) Both the statements (A) and (B) are incorrect

(3) Both the statements (A) and (B) are correct

(4) Only statement (B) is correct

150. Read the following properties of a tissue:

- Matrix contains collagen and mineral salts (Mostly calcium and phosphates)
- Possesses small fluid-filled spaces in concentric rings of extracellular matrix
- Houses blood forming tissue

Identify the tissue and select its correct location in humans.

(1) In the tip of nose

(2) In long bones of the leg

(3) Between adjacent bones of the vertebral column

(4) In the outer ear joints

151. Select the option that **correctly** identifies the difference between ribose and deoxyribose sugars.

	Ribose	Deoxyribose
(1)	Present in RNA; has –OH at C2'	Present in DNA; has –H at C2'
(2)	Pentose sugar found in DNA	Hexose sugar found in RNA
(3)	Has 4 oxygen atoms	Has 5 oxygen atoms
(4)	Is a disaccharide	Is a monosaccharide

152. **Assertion (A):** Biocatalysts in the human body are capable of accelerating biochemical reactions that would otherwise proceed extremely slowly at physiological temperature.

Reason (R): Enzymes function by providing an alternative reaction pathway with a lower activation energy and they increase the rate of reaction by being consumed in the process.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) correctly explains (A)

(2) Both (A) and (R) are true, but (R) does not explain (A) correctly

(3) (A) is true, (R) is false

(4) (A) is false, (R) is true

153. The common structural feature between cellulose and cytidylic acid is that both

(1) Contain phosphodiester bonds in their structure

(2) Are found exclusively in the acid -insoluble fraction of cell extracts

(3) Contain glycosidic bond

(4) Are branched polymers

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154. Select the **incorrect** match.

(1) Chitin - Homopolymer of N-acetylglucosamine

(2) Triglyceride - Composed of ether bonds

(3) GLUT-4 - Possesses peptide bonds

(4) Cholesterol - Does not contain fatty acids

155. Choose the **correct** statement.

(1) All the biomolecules found in the retentate fraction after trichloroacetic acid treatment are polymeric in nature.

(2) Ribozymes are RNA molecules with catalytic activity and thus, contain peptide bonds in their structure.

(3) All amino acids have a $-CH_3$ group as their R-group.

(4) The tertiary structure of proteins is stabilized by hydrogen bonds, ionic bonds and disulphide bonds.

156. Which among the following structures is ionizable in nature due to presence of $-NH_2$ and COOH groups and exist in zwitterionic form?

(1) Adenine

(2) Cytosine

(3) Valine

(4) Vinblastin

157. At the moist surface of buccal cavity, 'X' epithelium is present which primarily helps in

(1) Secretion and absorption

(2) Forming a diffusion boundary

(3) Moving particles or mucus in a specific direction over the epithelium

(4) Providing protection against chemical and mechanical stresses

158. A student was analysing the epithelium lining the two structures named PCT of nephron and small intestine. He found that the epithelia present in both the structures are similar to each other in

(a) Presence of microvilli at the apical surfaces

(b) Presence of round and centrally located nuclei

(c) Performing functions like secretion and absorption

(d) Presence of tall and slender cells

Select the **correct** option

(1) (a) and (c)

(2) (a) and (b)

(3) (c) and (d)

(4) (b) and (d)

159. Choose the **correct** match.

(1)	Neurons	-	Excitable and actively dividing cells of neural tissue
(2)	Muscle fibres	-	Shorten in response to stimulation
(3)	Blood	-	Contains erythrocytes, histiocytes and thrombocytes
(4)	Glandular epithelium	-	Possesses modified flattened and squamous cells

160. The total number of fatty acid molecules in lecithin is equal to the total number of oxygen atoms in the

(1) Purine present both in RNA and DNA

(2) Pyrimidine exclusively present in RNA

(3) Monomeric unit of starch

(4) Alcoholic amino acid

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161. Which of the following correctly describes the function of glycogen?

(1) It is present as a store house of energy in animal tissues.

(2) It is a structural heteropolymer, present in dense irregular connective tissue

(3) It is a storage homopolymer found exclusively in the liver

(4) It is a globular protein present on the surface of lymphocytes

162. Select the **correct** match w.r.t. secondary metabolites.

(1) Lectin - Abrin

(2) Toxin - Concanavalin A

(3) Drug - Anthocyanin

(4) Alkaloid - Codeine

163. Arachidonic acid is a polyunsaturated fatty acid, that contains 20 carbon atoms with 4 double bonds. Its molecular formula is

(1) $C_{20}H_{34}O_2$ (2) $C_{20}H_{32}O_2$

(3) $C_{20}H_{40}O_2$ (4) $C_{20}H_{30}O_4$

164. **Assertion (A):** When an egg is boiled, the egg white transforms from a liquid to a solid coagulated state, which is irreversible.

Reason (R): At temperature below the optimum value, the covalent interactions that maintain the 3D structure of proteins are disrupted, causing the polypeptide chain to unfold.

In the light of above statements, select the correct option.

(1) (A) is true, (R) is false

(2) Both (A) and (R) are true but (R) does not explain (A) correctly

(3) Both (A) and (R) are true and (R) correctly explains (A)

(4) Both (A) and (R) are false

165. Match Column I with Column II and select the correct option.

	Column I		Column II
a.	RuBisCO	(i)	Catalyses the formation of H_2CO_3
b.	Haemoglobin	(ii)	Secondary structure of protein
c.	Alpha helix	(iii)	Exhibits quaternary structure and aids in transport of gases
d.	Carbonic anhydrase	(iv)	Most abundant protein in whole of the biosphere

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iii), b(ii), c(iv), d(i)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(iv), b(iii), c(ii), d(i)

166. Connective tissues are generally characterised by all of the following, **except**

(1) Cells are widely separated by intercellular matrix

(2) Ground substance made of modified polysaccharides and proteins

(3) Cells having excitable plasma membranes capable of generating action potentials

(4) Unlike the epithelial tissue, it does not face the outside/external environment as epithelial tissue.

167. Choose the **odd** one w.r.t. locations that is characterised by the presence of muscle fibres that usually do not get fatigued and are uninucleated.

(1) Biceps

(2) Urinary bladder

(3) Stomach

(4) Iris of eye

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168. In humans, the epithelium lining the inner walls of blood vessels is best described as
(1) Transitional epithelium that stretches and recoils
(2) Simple squamous that allows diffusion
(3) Ciliated columnar that moves blood forward
(4) Simple cuboidal to aid in absorption

169. A histologist performs a study on a tissue from a body part exposed to constant mechanical wear and tear. The expected tissue type would be
(1) Simple squamous epithelium
(2) Glandular epithelium
(3) Simple columnar epithelium
(4) Compound epithelium

170. In a hypothetical condition, if A drug 'X' enhances the function of mast cells in the connective tissue, then the effect on the patients consuming this drug would show
(1) Increased blood clotting
(2) Improved phagocytosis by mast cell
(3) Increased secretion of serotonin and histamine
(4) Increased fat storage

171. _____ and _____ are the secretions produced by a unicellular gland and a multicellular gland in humans, respectively. Choose the option that fills the respective blanks correctly.
(1) Mucus, Saliva (2) Sweat, oil
(3) Insulin, Milk (4) Wax, Pepsin

172. Read the given statements and select the correct option.

Statement (A): The inhibitors that have structural similarities with the substrate are often used in the control of bacterial pathogens.

Statement (B): Once the enzymatic reaction reaches its maximum velocity, its V_{max} can be exceeded by further rise in concentration of the substrate.

(1) Both the statements (A) and (B) are correct
(2) Both the statements (A) and (B) are incorrect
(3) Only statement (B) is correct
(4) Only statement (A) is correct

173. A comparison was done between lysine and tyrosine. It was found that
(1) Former one possesses an extra carboxylic group
(2) Latter one possesses a cyclic ring in its structure
(3) Former one has equal number of amino and carboxylic groups
(4) Latter one possesses an additional amino group

174. Glycogen is similar to starch as both of them
(1) Contain glycosidic bond
(2) Are unbranched homopolymers
(3) Lack complex helices
(4) Lack non-reducing end

175. How many of the biomolecules listed in the box given below contains peptide bonds in its/their structure(s)?

Carboxypeptidase, Antibody, Receptor, Trypsin, Chitin

Select the correct option.
(1) One (2) Four
(3) Three (4) Two

176. A linear polysaccharide chain is formed of 'X' glucose units. If during the formation of this polysaccharide chain, 230 water molecules were removed, then the polysaccharide consists of how many 'X' glucose units?
(1) 230
(2) 229
(3) 232
(4) 231

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177. _____ are the most commonly found cells in loose connective tissue and they are responsible for synthesising and secreting the fibres and ground substance of the extracellular matrix.

Select the option that correctly fills the blank.

(1) Chondroblasts

(2) Osteoblasts

(3) Fibroblasts

(4) Plasma cells

178. Observe the given diagram of a protein molecule and choose the **incorrect** option w.r.t. it.



(1) Possesses hydrogen bonds in its structure

(2) Included in the same category of structure to which β-pleated sheet belongs

(3) Possesses peptide bond in its structure

(4) Active site of enzyme exclusively shows this structure

179. Consider the given reaction and select the correct option w.r.t. the class of enzyme that catalyses this reaction.

$S - G + S' \rightarrow S + S' - G$

(1) Catalyses the transfer of hydrogen from one substrate to another

(2) Involves the addition of water

(3) Belongs to class II of enzymes

(4) Catalyses the rearrangement of molecular structure to form isomers

180. Select the **incorrect** match.

(1)	Catalase	–	Possesses haem as the prosthetic group
(2)	Carboxypeptidase	–	Enzyme that contains zinc as the co-factor
(3)	Niacin	–	Present in the co-enzyme NADP
(4)	Peroxidase	–	Possesses the co-factor which is transiently bound to the holoenzyme

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